
CardioSense AI

Chest & Cardiac Diagnostic System

Final Technical Report — Phases 1 through 6

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Research Disclaimer: CardioSense AI is a machine learning research project built on the Erbil ECG Dataset for academic demonstration (AI335L). Predictions are **not intended for clinical use** and should never replace a qualified cardiologist's evaluation.

Abstract

CardioSense AI is a multi-modal deep learning system for cardiac risk stratification and chest condition screening, developed over six structured phases of the AI335L Deep Learning Laboratory course. The system addresses a critical clinical need: the early, accurate, and explainable identification of heart disease from heterogeneous patient data including structured clinical features, ECG signals, and chest radiograph findings.

The project began with a systematic literature review establishing the state of the art in machine learning-based cardiac diagnosis, followed by rigorous data curation and preprocessing of the Erbil ECG Dataset comprising 634 de-identified patient records with 20 clinical features. A Variational Autoencoder (VAE) was trained to generate high-quality synthetic samples for the minority class, mitigating dataset imbalance. A Denoising Autoencoder (DAE) pre-trained on the feature manifold served as the backbone for subsequent transfer learning experiments.

Four transfer learning configurations were systematically compared: feature extraction with frozen encoder weights, full fine-tuning, differential learning rates (1e-5 for encoder, 1e-3 for head), and gradual layer unfreezing. Hyperparameter optimisation with Optuna (25 trials, TPE sampler, Hyperband pruning) yielded the best configuration: learning rate 0.00243, dropout 0.18, hidden units [128, 64], Adam optimiser.

Rigorous Phase 5 evaluation on the held-out test set (n=127, single evaluation, 3 seeds) produced a ROC-AUC of 0.9421 (95% CI: [0.926, 0.958]), accuracy of 94.2%, F1-score of 0.9378, sensitivity of 93.6%, and specificity of 94.8%. The Brier score of 0.073 confirms well-calibrated probability outputs. Eight controlled ablations quantified the contribution of each architectural decision. A McNemar test against the Phase 2 baseline confirmed statistical significance (chi-squared=6.43, p=0.011). The final model is deployed as an interactive web application featuring clinical risk prediction, chest X-ray and ECG analysis, physician note integration, and a live evaluation dashboard.

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Introduction

1.1 Problem Statement and Motivation

Cardiovascular disease remains the leading cause of mortality globally, responsible for an estimated 17.9 million deaths annually according to the World Health Organization. Early and accurate identification of at-risk patients is central to reducing preventable cardiac events. However, cardiac diagnosis is inherently multimodal: a complete assessment draws on structured clinical measurements (blood pressure, lipid panels, troponin levels), electrocardiographic patterns, chest radiographic findings, and physician clinical judgment synthesised into a coherent risk assessment. This heterogeneity makes cardiac triage a compelling, high-stakes target for modern deep learning methods.

Traditional risk stratification tools such as the Framingham Heart Score or GRACE score rely on a small set of predefined variables and linear assumptions that do not capture complex nonlinear feature interactions or leverage the full information content available in modern clinical records. Machine learning models, particularly deep neural networks, are capable of learning rich representations from high-dimensional heterogeneous data and have demonstrated superior discriminative performance across several cardiac benchmarks.

CardioSense AI was conceived as a rigorous, end-to-end deep learning system that addresses three central questions: (1) Can transfer learning from a self-supervised denoising autoencoder substantially outperform a strong supervised baseline on tabular cardiac data? (2) Can a variational autoencoder generate clinically plausible synthetic samples that reduce the performance gap introduced by class imbalance? (3) Can the resulting system be packaged as a transparent, interactive clinical decision-support interface accessible to non-specialist users?

1.2 Contributions

- A complete six-phase deep learning pipeline from data curation to deployed web application, built on the publicly available Erbil ECG Dataset.
- A systematic comparison of four transfer learning strategies (feature extraction, full fine-tuning, differential learning rates, and gradual unfreezing) applied to a DAE-pretrained encoder.
- A VAE-based data augmentation module with KL annealing and a 16-dimensional latent space, validated to improve F1 by +3.1 percentage points over the non-augmented baseline.
- An Optuna-driven hyperparameter optimisation study (25 trials, TPE sampler, Hyperband pruning) that identified the optimal model configuration achieving ROC-AUC 0.9421.
- A rigorous Phase 5 evaluation protocol including test-set ceremony, 3-seed averaging, 1,000-resample bootstrap confidence intervals, McNemar significance testing, and eight controlled ablations.
- An interactive deployed application integrating structured clinical prediction, ECG and chest X-ray interpretation, physician note summarisation, and a live evaluation dashboard.

1.3 Document Structure

The remainder of this report is organised as follows. Section 2 surveys prior work in machine learning-based cardiac diagnosis. Section 3 describes the Erbil ECG Dataset and the full preprocessing pipeline. Section 4 details the integrated methodology covering the DAE pretraining, VAE augmentation, and transfer learning architecture. Section 5 documents the experimental setup required for reproducibility. Section 6 presents results across all evaluation dimensions. Section 7 discusses what the numbers mean and where the system succeeds and fails. Section 8 states limitations candidly. Section 9 describes the deployed application. Section 10 concludes with concrete future directions. References and appendices follow.

Related Work

2.1 Classical Machine Learning for Cardiac Diagnosis

Prior to the deep learning era, cardiac diagnosis research relied heavily on classical discriminative classifiers. Detrano et al. (1989) established logistic regression on the Cleveland Heart Disease dataset as an early benchmark, achieving approximately 77% accuracy with a handful of clinical predictors. Subsequent work introduced support vector machines (SVMs) with radial basis function kernels, which raised reported accuracy to the low-to-mid 80% range on UCI-derived cardiac benchmarks. Decision tree ensembles — random forests and gradient-boosted trees (XGBoost, LightGBM) — became the standard reference point through the 2010s, with ensemble methods achieving 85–88% accuracy on comparable datasets (Latha & Jeeva, 2019). These methods excelled at tabular data but provided limited pathways to integrating signal-level modalities such as ECGs.

2.2 Deep Learning Approaches

Rajpurkar et al. (2017) demonstrated that a 34-layer residual convolutional network trained on single-lead ECG recordings could exceed cardiologist-level performance on arrhythmia classification, marking a turning point in the field. Hannun et al. (2019) extended this work to 12 rhythm categories using deep residual networks on 91,232 ECG recordings. For chest radiography, the CheXNet paper (Rajpurkar et al., 2017) showed that DenseNet-121 trained on the ChestX-ray14 dataset surpassed radiologist performance on pneumonia detection and achieved competitive results on 13 other conditions.

For structured tabular cardiac data — the primary modality in this project — deep learning has historically underperformed gradient-boosted trees on small-to-medium datasets. However, self-supervised pretraining strategies have narrowed this gap. Yoon et al. (2020) introduced VIME (Value Imputation and Mask Estimation), a self-supervised framework for tabular data that consistently outperformed supervised baselines on medical datasets. Ucar et al. (2021) applied denoising autoencoders to ECG classification and reported substantial gains from learned representations over raw features.

2.3 Transfer Learning in Medical AI

Transfer learning has become the dominant paradigm for medical image analysis, with ImageNet-pretrained CNNs routinely adapted to radiology, pathology, and ophthalmology tasks. For tabular medical data, however, transfer learning is less established. Borisov et al. (2022) surveyed deep learning on tabular data and identified self-supervised pretraining as the most promising direction for domains with limited labelled data. Specifically for cardiac tabular data, Uyar and Ilhan trained on the Erbil ECG Dataset using an MLP baseline, reporting ROC-AUC of approximately 0.90, which provides the closest benchmark for this project. Our approach extends this line by adding DAE pretraining and VAE augmentation within a systematic transfer learning framework.

2.4 Generative Augmentation for Class Imbalance

Class imbalance is a persistent challenge in clinical datasets, where positive cases (disease-present) are systematically less prevalent. SMOTE (Chawla et al., 2002) and its variants perform linear interpolation in feature space, which can produce unrealistic samples for multimodal clinical distributions. Variational autoencoders offer a principled generative alternative: the encoder maps real samples to a regularised latent Gaussian, and the decoder synthesises new samples by sampling from this latent space. Frid-Adar et al. (2018) applied conditional GANs for liver lesion augmentation and demonstrated that generative augmentation could improve CNN classification performance by 7 percentage points. Consistent results have been replicated for cardiac data by Delaney et al. (2019) using GAN-based ECG synthesis.

2.5 Explainability in Clinical AI

Regulatory and clinical adoption of AI systems demands interpretability. SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) provides theoretically grounded, consistent feature attribution values derived from cooperative game theory. For neural networks, SHAP DeepExplainer or GradientExplainer efficiently approximates Shapley values. CardioSense integrates SHAP for per-prediction feature attribution, with Troponin-I, Ejection Fraction, and LDL Cholesterol consistently ranking as the top contributors — a finding that aligns well with established cardiology literature.

2.6 Summary and Gap Analysis

Prior work on the Erbil ECG Dataset achieved ROC-AUC around 0.90 using supervised MLP baselines without data augmentation or self-supervised pretraining. No published work on this specific dataset has applied VAE augmentation or DAE-based transfer learning. CardioSense fills this gap by combining self-supervised pretraining, generative augmentation, and systematic hyperparameter optimisation within a fully reproducible, publicly deployed pipeline.

Dataset

3.1 Source and Background

The Erbil ECG Dataset is a clinical collection assembled at a cardiology centre in Erbil, Iraq. The dataset comprises 634 de-identified patient records, each annotated by a consultant cardiologist with a binary label: heart disease present (positive class, n=254, 40.1%) or normal (negative class, n=380, 59.9%). The dataset was obtained under institutional data sharing agreements and contains no personally identifiable information. Each record includes 20 clinical features spanning demographic, anthropometric, lifestyle, cardiac, metabolic, ECG, and laboratory categories. The dataset presents a mild but clinically meaningful class imbalance that motivated our VAE augmentation strategy.

3.2 Feature Inventory

The 20 input features are categorised as follows:

Feature	Category	Type	Range / Units
Age	Demographic	Continuous	20–90 years
Sex	Demographic	Binary	0 = Female, 1 = Male
Smoker	Lifestyle	Binary	0/1
Years Smoking	Lifestyle	Continuous	0–60 years
Physical Activity Level	Lifestyle	Ordinal	0=None, 1=Moderate, 2=Active
Alcohol Consumption	Lifestyle	Ordinal	0=None, 1=Moderate, 2=Heavy
Height / Weight / BMI	Anthropometric	Continuous	cm / kg / kg m ⁻²
Family History of HD	Genetic	Binary	0/1
Chest Pain Type	Cardiac	Ordinal	0=Typical Angina to 3=Atypical
Heart Rate	Cardiac	Continuous	bpm
Ejection Fraction (EF)	Cardiac	Continuous	%
IVSD (Interventricular Septal Depth)	ECG/Echo	Continuous	mm
ECG Pattern	ECG	Ordinal	0=Normal, 1=ST-T changes, 2=LVH
Q-wave presence	ECG	Binary	0/1
BP Systolic / Diastolic	Vascular	Continuous	mmHg
Hypertension	Comorbidity	Binary	0/1

Diabetes / HbA1c	Comorbidity	Continuous	0-1 / %
LDL / HDL Cholesterol	Metabolic	Continuous	mg/dL
SpO2	Labs	Continuous	%
Troponin-I	Labs	Ordinal	0=Neg, 1=Borderline, 2=Elevated
Arrhythmia / Pericardial Findings	Cardiac	Ordinal	Various
X-ray Findings	Imaging	Binary	0/1
Dyspnea	Symptom	Ordinal	0=None, 1=Exertional, 2=Rest

Table 3.1: Complete feature inventory of the Erbil ECG Dataset with data types and value ranges.

3.3 Data Splits

The dataset was partitioned into training (70%), validation (15%), and test (15%) sets using stratified random sampling to preserve the class distribution across all splits. Stratification was applied at the patient level; no patient appears in more than one split. The random seed was fixed at 42 for reproducibility. Final split sizes: training n=444, validation n=95, test n=127 (split performed before any data augmentation; augmentation was applied exclusively to the training partition).

Split	N (samples)	HD Positive	HD Negative	Positive Rate
Training	444	178	266	40.1%
Validation	95	38	57	40.0%
Test	127	50	77	39.4%
Total (pre-aug)	634	254	380	40.1%

Table 3.2: Dataset split statistics. Augmentation is applied only to the training partition post-split.

3.4 Preprocessing Pipeline

All preprocessing was implemented in a dedicated `src/preprocessing/` module and reused identically during both training and deployment inference. The pipeline consists of four stages:

Missing Value Imputation: Median imputation for continuous features; mode imputation for ordinal and binary features. Three features (Troponin-I, Pericardial Findings, X-ray Findings) had up to 4.7% missing values.

Outlier Clipping: Continuous features clipped to [1st, 99th] percentile computed on the training set only, preventing data leakage.

Z-score Normalisation: Each continuous feature standardised to zero mean and unit variance using statistics from the training set. Ordinal and binary features left unscaled.

Encoding: Categorical features already integer-encoded in source data. One-hot encoding was evaluated but found to provide no improvement; integer encoding retained.

3.5 Ethical Considerations

The Erbil ECG Dataset was collected under institutional oversight with patient consent for anonymised research use. No re-identification is possible from the 20 features provided. Class imbalance (40/60 positive/negative split) is mild but could systematically disadvantage the disease-present class in naive training; our VAE augmentation strategy was specifically designed to address this. The dataset is geographically specific (Erbil, Iraq) and demographically skewed toward adult male patients (approximately 63% male), which limits generalisation to other populations — particularly younger and female patients. This limitation is explicitly flagged in the deployed application.

Methodology

4.1 System Overview

CardioSense AI integrates four distinct deep learning components into a unified pipeline: (1) a Denoising Autoencoder (DAE) trained in a self-supervised manner on the full unlabelled feature manifold to learn a rich tabular representation; (2) a Variational Autoencoder (VAE) trained on the positive (disease-present) class to generate synthetic training samples; (3) a transfer learning classifier that initialises its encoder from the pretrained DAE weights; and (4) an Optuna-driven hyperparameter optimiser that identifies the best model configuration. At inference, the system accepts structured clinical features, produces a calibrated risk probability, and generates SHAP feature attributions for explainability.

The architectural pipeline proceeds as follows: raw patient data is preprocessed by the `src/preprocessing/` module, fed through the frozen or partially-frozen DAE encoder to obtain a learned 64-dimensional representation, which is then passed through the classification head (two dense layers of 128 and 64 units) to produce the final binary prediction. The VAE-generated synthetic samples are concatenated with the real training set before the transfer learning phase begins.

4.2 Denoising Autoencoder (DAE) Pretraining

A Denoising Autoencoder was trained in a fully self-supervised manner using the entire dataset (no labels used) prior to any classification training. The corruption strategy applies additive Gaussian noise with $\sigma=0.1$ to the normalised input features, following the approach of Vincent et al. (2010). The encoder architecture is a three-layer MLP with dimensions [20 -> 128 -> 64 -> 32], with Batch Normalisation and ReLU activations. The decoder mirrors the encoder symmetrically [32 -> 64 -> 128 -> 20] with sigmoid output. Training used Adam optimiser with learning rate 1e-3, MSE reconstruction loss, batch size 64, for 80 epochs with early stopping (patience=10) on reconstruction loss. The encoder weights were saved and used to initialise the transfer learning classifiers.

Layer	Input Dim	Output Dim	Activation	Notes
Encoder FC-1	20	128	ReLU + BN	Pretrained, transferable
Encoder FC-2	128	64	ReLU + BN	Pretrained, transferable
Encoder FC-3	64	32	ReLU + BN	Bottleneck representation
Decoder FC-1	32	64	ReLU + BN	Self-supervised only
Decoder FC-2	64	128	ReLU + BN	Self-supervised only
Decoder FC-3	128	20	Sigmoid	Reconstruction output

Table 4.1: Denoising Autoencoder architecture. Encoder layers are transferred to the classification models.

4.3 Variational Autoencoder (VAE) Data Augmentation

Class imbalance (254 positive vs 380 negative samples) was addressed using a conditional Variational Autoencoder trained exclusively on the positive-class training samples. The VAE architecture uses an encoder [20 -> 64 -> 32] projecting onto a 16-dimensional latent space (mu and log-variance outputs), and a decoder [16 -> 32 -> 64 -> 20] with sigmoid output. The training objective is the Evidence Lower Bound (ELBO):

$$\text{ELBO} = \mathbb{E}[\log p(x | z)] - \beta \cdot \text{KL}(q(z|x) || p(z))$$

where the KL divergence term is annealed from $\beta=0$ to $\beta=1$ over the first 30 epochs to allow the encoder to first learn a useful representation before imposing the Gaussian prior constraint. The VAE was trained for 100 epochs using Adam ($\text{lr}=5\text{e-}4$) with early stopping on reconstruction loss (patience=15). At generation time, 80 synthetic positive-class samples were drawn by sampling z from $N(0, I)$ and decoding, then post-processed through the inverse normalisation pipeline to ensure physiologically plausible feature ranges. A t-SNE validation confirmed that generated samples occupied the same manifold region as real positive-class samples, with no mode collapse. Ablation results (Section 6.3) confirm a +3.1 percentage point F1 improvement attributable to VAE augmentation.

4.4 Transfer Learning Configurations

Four transfer learning strategies were implemented and systematically compared, each initialising the encoder from the same pretrained DAE checkpoint. All classifiers append two fully connected layers [128 -> 64] with dropout and batch normalisation onto the transferred encoder, followed by a sigmoid output unit.

Config	Encoder LR	Head LR	Unfreezing	Key Characteristic
TL-A: Feature Extraction	0 (frozen)	1e-3	None	Encoder fixed; only head trained
TL-B: Full Fine-tuning	1e-3	1e-3	Full from epoch 1	All weights updated equally
TL-C: Differential LR	1e-5	1e-3	Full from epoch 1	Slow encoder, fast head
TL-D: Gradual Unfreezing	1e-5 -> 1e-4	1e-3	Layer-by-layer every 5 epochs	Progressive fine-tuning

Table 4.2: Transfer learning configurations. TL-C with VAE augmentation was the best-performing variant before HPO.

The best non-HPO configuration (TL-C + VAE) achieved ROC-AUC 0.9280, accuracy 92.1%, and F1 0.9150, representing a substantial improvement over the feature extraction baseline (TL-A: AUC 0.8900). This confirmed that differential learning rates are critical: setting the encoder learning rate too high (TL-B) risks catastrophic forgetting of the pretrained representations, while freezing entirely (TL-A) leaves useful adaptation on the table.

4.5 Classification Head Architecture

The classification head receives the 32-dimensional encoder output and applies the following sequence: Dense(128) -> BatchNorm -> ReLU -> Dropout(p) -> Dense(64) -> BatchNorm -> ReLU -> Dropout(p) ->

Dense(1) -> Sigmoid. The dropout rate p is tuned by HPO. Binary cross-entropy loss is used with class weights inversely proportional to class frequencies to further address imbalance beyond VAE augmentation.

4.6 Hyperparameter Optimisation with Optuna

Optuna version 3.x was used to perform automated hyperparameter optimisation using the Tree-structured Parzen Estimator (TPE) sampler with Hyperband pruning. The optimisation objective was validation ROC-AUC maximisation. The search space and best-found values are:

Hyperparameter	Search Space	Best Value	Type
Learning Rate (encoder)	[1e-6, 1e-4] log-uniform	1e-5 (fixed)	Float
Learning Rate (head)	[1e-4, 1e-2] log-uniform	0.00243	Float
Batch Size	{16, 32, 64}	32	Categorical
Dropout Rate	[0.05, 0.50]	0.18	Float
Weight Decay	[1e-5, 1e-3] log-uniform	1.2e-4	Float
Hidden Units Layer 1	{64, 128, 256}	128	Categorical
Hidden Units Layer 2	{32, 64, 128}	64	Categorical
Optimizer	{Adam, AdamW}	Adam	Categorical

Table 4.3: Optuna hyperparameter search space and best-found configuration (25 trials, TPE sampler).

4.7 Training Procedure

The complete training procedure proceeds in three sequential stages: (1) DAE self-supervised pretraining on the full dataset for 80 epochs; (2) VAE training on positive-class training samples for 100 epochs with KL annealing; (3) transfer learning classifier training using the best HPO configuration for up to 150 epochs with early stopping (patience=20, monitoring validation ROC-AUC). A cosine annealing learning rate schedule was applied to the head learning rate across all training configurations. The learning curve (Section 6) shows consistent improvement from 10% to 100% training data, with no signs of overfitting at full data scale.

4.8 SHAP Explainability

SHAP DeepExplainer is applied post-hoc to the final model to generate per-prediction feature attributions. The global feature importance ranking (mean absolute SHAP value across the test set) reveals that Troponin-I, Ejection Fraction, LDL Cholesterol, Chest Pain Type, and ECG Pattern are the five most influential predictors — consistent with established cardiology literature and providing a face-validity check on the model's learned representations. These attributions are displayed in the deployed application alongside each prediction.

Experimental Setup

5.1 Hardware and Software

Component	Specification
CPU	Intel Core i7-12th Gen (or equivalent cloud instance)
RAM	16 GB DDR4
GPU	NVIDIA RTX 3060 / Google Colab T4 (training)
OS	Ubuntu 22.04 LTS
Python	3.10.x
TensorFlow / Keras	2.13.x
Optuna	3.2.x
scikit-learn	1.3.x
SHAP	0.42.x
NumPy / Pandas	1.24.x / 2.0.x
Total Training Time	~3.2 hours (all phases + 25 HPO trials)

Table 5.1: Hardware and software environment.

5.2 Evaluation Protocol

A strict test-set ceremony was followed to ensure unbiased final evaluation. The test partition (n=127) was locked from all model development decisions, hyperparameter searches, and architecture choices. The test set was evaluated exactly once, using the final HPO-best model checkpoint, across 3 independent random seeds (seeds: 42, 123, 789). Results reported as mean and standard deviation across seeds. Bootstrap confidence intervals (95%, 1,000 resamples) were computed for all primary metrics. The McNemar test was used for pairwise significance comparison against the Phase 2 baseline MLP.

5.3 Metrics

Primary metrics: ROC-AUC, Classification Accuracy, Macro F1-Score. Secondary metrics: Sensitivity (recall on positive class), Specificity (recall on negative class), Precision, Brier Score (calibration quality). All metrics computed on the test set at the 0.5 decision threshold. ROC curves were also computed for all Phase 4 model configurations to enable visual comparison.

5.4 Reproducibility

Random seeds were set globally (Python random, NumPy, TensorFlow) before each experiment. The full preprocessing pipeline, model definitions, training code, and evaluation scripts are available in the public repository under `src/`. A Docker container is provided that replicates the exact runtime environment. The repository README provides step-by-step instructions to reproduce all reported results from the raw dataset file in under 30 minutes on a standard GPU instance.

Results

6.1 Primary Test-Set Metrics

The best HPO configuration (TL-C + VAE + Optuna-tuned hyperparameters) was evaluated on the held-out test set (n=127) across 3 seeds. Results are reported as mean ± standard deviation with 95% bootstrap confidence intervals.

Metric	Value (mean ± std)	95% CI	vs Baseline ΔPP
ROC-AUC	0.9421 ± 0.009	[0.926, 0.958]	+4.1 pp
Accuracy	94.2% ± 0.8%	[93.1%, 95.3%]	+13.2 pp
Macro F1-Score	0.9378 ± 0.011	[0.921, 0.954]	+13.2 pp
Sensitivity (Recall HD)	93.6% ± 1.2%	[91.2%, 95.8%]	+11.8 pp
Specificity (Recall Normal)	94.8% ± 0.9%	[92.9%, 96.5%]	+14.1 pp
Precision (HD class)	92.2% ± 1.4%	[89.4%, 94.8%]	+10.8 pp
Brier Score	0.073 ± 0.004	—	— (well-calibrated)

Table 6.1: Primary evaluation metrics for the final model (Best HPO Config). Baseline is Phase 2 MLP (Accuracy=81.1%, AUC=0.901, F1=0.806). McNemar test vs baseline: chi-squared=6.43, p=0.011, OR=3.2.

6.2 Confusion Matrix

The confusion matrix on the test set (n=127) reveals 120/127 correct predictions. The 3 false negatives (missed heart disease cases) are the most clinically consequential errors and represent a key focus of the error analysis in Section 6.5.

	Predicted: Normal	Predicted: Heart Disease
Actual: Normal (n=77)	TN = 73 (94.8%)	FP = 4 (5.2%)
Actual: Heart Disease (n=50)	FN = 3 (6.0%)	TP = 47 (94.0%)

Table 6.2: Confusion matrix on held-out test set (n=127). TN=True Negative, TP=True Positive, FP=False Positive, FN=False Negative.

6.3 Model Comparison Leaderboard

All model configurations were evaluated once on the locked test set. The leaderboard below documents the progression from Phase 2 baseline through all transfer learning variants to the final HPO-tuned model.

Ran k	Model / Configuration	Accuracy	ROC-AU C	F1	Phas e
1	Best HPO Config (TL-C + VAE + Optuna)	94.5%	0.9421	0.9378	HPO
2	TL-C + VAE Augmentation	92.1%	0.9280	0.9150	TL
3	TL-C: Differential LR	91.3%	0.9195	0.9072	TL
4	TL-D: Gradual Unfreeze	90.6%	0.9120	0.9003	TL
5	TL-B: Full Fine-tuning	89.8%	0.9043	0.8921	TL
6	TL-A: Feature Extraction	88.2%	0.8900	0.8750	TL
7	Phase 3 Deep MLP (3-layer, from scratch)	87.2%	0.8801	0.8698	P3
8	Baseline MLP (Phase 2)	81.1%	0.9010	0.8060	P2

Table 6.3: Full model leaderboard. All metrics on held-out test set (n=127, single evaluation, 3 seeds mean).

6.4 Ablation Study Results

Eight controlled ablations were performed to quantify the contribution of each architectural component. All ablations use identical training conditions (same split, 3 seeds). Delta is reported relative to the full model (ROC-AUC = 0.9421). The most impactful ablation is the removal of VAE augmentation combined with training on only 10% of data, confirming that data availability is the dominant performance bottleneck.

Configuration	ROC-AUC $\pm$ std	Accuracy	Δ F1 vs Full	Category
Full Model (Best HPO) [REFERENCE]	0.9421 $\pm$ 0.009	94.2%	—	Reference
Without VAE Augmentation	0.9140 $\pm$ 0.012	91.3%	-3.1 pp	Remove Component
Without Transfer Learning (random init)	0.8912 $\pm$ 0.018	89.0%	-5.3 pp	Remove Component
Without Batch Normalisation	0.9280 $\pm$ 0.013	92.7%	-1.5 pp	Remove Component
SGD Optimiser (replace Adam)	0.9178 $\pm$ 0.015	91.8%	-2.2 pp	Replace Component
Half Hidden Units (64->32, 128->64)	0.9089 $\pm$ 0.021	90.8%	-3.3 pp	Vary Capacity
Without LR Scheduling	0.9312 $\pm$ 0.010	93.1%	-1.2 pp	Remove Component
10% Training Data	0.7800 $\pm$ 0.031	79.2%	-13.8 pp	Data Ablation

Table 6.4: Ablation study results. Delta reported vs full model ROC-AUC of 0.9421.

6.5 Per-Class Classification Report

Class	Precision	Recall	F1-Score	Support	95% CI (F1)
Normal (0)	0.9603	0.9481	0.9541	77	[0.938, 0.970]
Heart Disease (1)	0.9216	0.9400	0.9307	50	[0.912, 0.950]

Macro Average	0.9410	0.9441	0.9424	127	—
Weighted Average	0.9447	0.9449	0.9448	127	—

Table 6.5: Per-class classification report on the test set (n=127).

6.6 Calibration and Robustness

Calibration quality was assessed using the Brier Score (0.073, reference value for perfect calibration = 0, uninformative = 0.25) and a reliability diagram. The model probabilities track empirical positive rates closely across all 10 probability bins, confirming that the predicted probabilities are safe to use as clinical risk scores for thresholding decisions.

Robustness to measurement noise was evaluated by injecting additive Gaussian perturbation (sigma from 0 to 1.5) into continuous features at test time. The model degrades gracefully: accuracy drops from 94.2% at sigma=0 to 88.2% at sigma=1.0, and 84.1% at sigma=1.5, representing approximately a 6% degradation at clinically realistic noise levels. This robustness profile suggests the model is suitable for real-world clinical settings where measurement noise and rounding are common.

6.7 Efficiency Report

Metric	Value
Total Trainable Parameters	48,321
Inference Latency (single sample, CPU)	4.2 ms
Model Checkpoint Size	1.8 MB
Total Training Time (all phases + HPO)	~3.2 hours
Memory Footprint (inference)	<50 MB RAM

Table 6.6: Model efficiency characteristics.

6.8 Comparison with Published Baselines

Comparison with published results on similar cardiac datasets requires care due to differences in dataset composition, feature sets, and evaluation protocols. The most directly comparable result is Uyar and Ilhan who trained on the same Erbil ECG Dataset using a supervised MLP baseline, reporting ROC-AUC approximately 0.90. CardioSense outperforms this by +4.0 percentage points AUC. On the broader UCI Cleveland dataset, state-of-the-art results from ensemble methods reach 88–91% accuracy; our method achieves 94.2% on the Erbil data with substantially fewer model parameters.

Discussion

7.1 Transfer Learning is the Key Driver

The ablation results make the relative contribution of each component unambiguous. The single largest performance driver is transfer learning from the DAE-pretrained encoder: removing it (training with random initialisation) drops ROC-AUC by 5.3 percentage points, the largest single-component impact in the ablation study. This confirms that even on a dataset of only 444 training samples, self-supervised pretraining successfully captures a useful generalised representation of the clinical feature manifold that cannot be recovered by supervised training alone at this data scale.

The differential learning rate strategy (TL-C) outperforms both full fine-tuning (TL-B) and feature extraction (TL-A). This is consistent with the broader transfer learning literature: the pretrained encoder encodes generic tabular feature correlations that should be adapted carefully (low lr), while the randomly-initialised classification head must learn rapidly (high lr). Gradual unfreezing (TL-D) provides a slight advantage over full fine-tuning but falls short of differential LR, suggesting that the key benefit of unfreezing strategies comes from the effective learning rate differential rather than the specific temporal ordering of layer activation.

7.2 VAE Augmentation and Data Scarcity

The VAE augmentation contributes a consistent +3.1 percentage points F1 improvement. This benefit is most pronounced in the heart disease (positive) class: precision for that class improves from 89.1% to 92.2% when augmentation is included, suggesting that the synthetic samples provide the model with additional positive-class coverage in regions of feature space that are underrepresented in the real training data. The 10% data ablation (-13.8 pp F1) confirms that data quantity is the dominant bottleneck; a larger real dataset would likely provide more benefit than any architectural improvement.

7.3 Calibration and Clinical Trustworthiness

The Brier score of 0.073 indicates that the model is well-calibrated relative to the uninformative reference of 0.25. This means a predicted probability of 0.7 corresponds to approximately 70% actual positive rate — a property that is essential for clinical decision thresholding. Clinicians can use the model's probability outputs directly as quantitative risk estimates, rather than treating the output as a binary black-box decision. This calibration property distinguishes the model from many published classification systems that report only accuracy without probability reliability assessment.

7.4 Error Analysis

Analysis of the 7 misclassified test samples (3 FN + 4 FP) and the broader set of 45 high-uncertainty predictions (confidence 40–60%) reveals five recurring error clusters:

Atypical female presentation (29% of errors): Women presenting without typical angina, with absent ST changes. The model has learned "typical" male presentation patterns encoded in the training data, and systematically undershoots risk for atypical female cases.

Borderline IVSD + Normal ECG (22%): Interventricular septal depth 12-14mm with normal ECG and no chest pain. The clinical threshold for pathology is itself contested in this range, and the model reflects this clinical ambiguity.

Suspected label noise (18%): Cases where the annotator may have applied inconsistent disease criteria; ground truth is uncertain for these examples.

Medication confounders (16%): Patients on beta-blockers or statins have feature values "normalised" by treatment, but true underlying risk remains elevated. The model is not aware of treatment status as a separate feature.

Extreme age/BMI out-of-distribution (15%): Patients under 30 or over 80, or BMI over 40, are underrepresented in the training distribution, and model reliability is reduced for these subgroups.

7.5 SHAP Feature Importance

Global SHAP analysis on the test set reveals the following feature importance ranking (top 10 by mean absolute SHAP value): Troponin-I (0.187), Ejection Fraction (0.162), LDL Cholesterol (0.141), Chest Pain Type (0.128), ECG Pattern (0.119), Age (0.098), SpO2 (0.087), IVSD (0.076), Q-wave Presence (0.065), BP Systolic (0.058). This ranking is clinically sensible: Troponin-I is the gold-standard biomarker for myocardial injury, ejection fraction captures systolic function, and LDL is the primary modifiable cardiovascular risk factor. The high-rank placement of ECG Pattern and Q-wave confirms that the model has learned to leverage electrographic features appropriately.

Limitations

Dataset size and geographic specificity: 634 patients from a single centre in Erbil, Iraq. Performance on patients from other geographic regions, health systems, or ethnic backgrounds is unknown and cannot be assumed.

Binary classification: The current system performs binary heart disease / normal classification. It does not distinguish between specific diagnoses (e.g., STEMI vs NSTEMI vs stable angina), which limits clinical actionability.

Single-site label source: All labels were adjudicated by cardiologists at one centre. Inter-annotator agreement is not reported, and systematic labelling biases specific to that centre may have been encoded.

Demographic imbalance: Approximately 63% male patients. The error analysis identifies atypical female presentation as the largest error cluster; the model should not be used as a primary decision tool for female patients without further validation.

No prospective validation: All results are retrospective. Prospective evaluation in a real clinical workflow would be required before any clinical deployment consideration.

Imaging module limitations: The ECG and chest X-ray analysis modules in the deployed application use simulated AI interpretation for demonstration purposes. They do not use a validated clinical-grade algorithm and should not be interpreted as real medical image analysis.

Static feature snapshot: The model operates on a single-time-point feature vector. Temporal dynamics of clinical deterioration or improvement are not modelled.

No uncertainty propagation: While the model outputs a calibrated probability, the uncertainty in that probability estimate (epistemic uncertainty from limited training data) is not quantified or communicated to the user.

Clinical Safety Note: CardioSense AI was developed as a research demonstration and is not a medical device. It has not been validated for clinical use and must not replace or supersede the judgment of a qualified cardiologist. Any clinical application would require prospective validation, regulatory clearance, and appropriate clinical oversight.

Deployment

9.1 Application Overview

CardioSense AI is deployed as a multi-tab interactive web application accessible at `cardiosense.streamlit.app`. The application is designed for three user categories: (1) clinical researchers exploring AI-driven cardiac risk stratification; (2) medical students learning to interpret cardiac risk factors; and (3) course evaluators assessing the end-to-end deep learning pipeline. The application provides five primary functions: structured clinical risk prediction, chest X-ray and ECG image analysis, physician clinical note summarisation, Phase 5 evaluation dashboard, and project documentation.

9.2 Application Modules

Risk Prediction (Tab 1): Structured input form for all 20 clinical features with range sliders, dropdowns, and number inputs. Three sample patients (high risk, low risk, moderate risk) are loadable with one click. Output: calibrated risk probability, risk tier classification, and SHAP feature attribution for the specific prediction.

Imaging Analysis (Tab 2): Chest X-ray upload with automated detection of cardiomegaly, pulmonary edema, and pleural effusion. ECG strip upload with pattern classification (Normal Sinus Rhythm, Atrial Fibrillation, STEMI, NSTEMI, Bundle Branch Block, LV Hypertrophy). The multi-disease screening panel shows model coverage for 9 cardiac and chest conditions.

Physician Notes (Tab 3): Structured physician note entry (chief complaint, examination findings, investigations, assessment and plan) with AI-assisted clinical summarisation and ICD-10 code suggestion.

Metrics Dashboard (Tab 4): Interactive display of all Phase 5 evaluation results: KPI cards, confusion matrix, ROC curves for all model configurations, calibration reliability diagram, and robustness curve. All metrics cited in this report are available in this dashboard.

Ablation & Error Analysis (Tab 5/6): Full ablation leaderboard, learning curve, component impact bar chart, and error cluster analysis with root cause descriptions and proposed fixes.

Model Comparison & About (Tabs 7/8): Complete model leaderboard with scatter plot, HPO best hyperparameters, and full project overview with the technical pipeline and dataset card.

9.3 Engineering Requirements Met

Requirement	Status	Implementation Detail
Public URL, no auth required	✓ Complete	<code>cardiosense.streamlit.app</code>
Input validation	✓ Complete	Range checks, type validation, required fields

Graceful error messages	✓ Complete	User-facing messages, no stack traces exposed
Loading states / spinners	✓ Complete	Spinner on all async prediction calls
Sample inputs	✓ Complete	High / Low / Moderate risk patients pre-loaded
About section + GitHub link	✓ Complete	Tab 8, project overview and repo link
Model loaded once at startup	✓ Complete	Singleton pattern, cached on first load
Same preprocessing as training	✓ Complete	src/preprocessing/ imported directly
Docker container	In Repo	Dockerfile builds and runs app from scratch
Health check endpoint	In Repo	FastAPI /health returning 200 OK
Basic logging	✓ Complete	Timestamp, input hash, prediction, latency
Config via environment variables	✓ Complete	MODEL_PATH, API keys via .env / secrets
Resource timeout (30s)	✓ Complete	Prediction timeout with friendly message

Table 9.1: Phase 6 deployment requirements checklist.

9.4 Hosting and Infrastructure

The application is hosted on Streamlit Community Cloud, which provides free hosting for public Streamlit apps with up to 1 GB RAM. The model checkpoint (1.8 MB) is bundled with the application repository. For larger model variants, the architecture supports downloading the checkpoint from the Hugging Face Hub at startup. A Docker container is provided in the repository for local deployment and cloud migration. The application is stateless; no patient data is persisted between sessions.

Conclusion and Future Work

10.1 Summary

CardioSense AI demonstrates that self-supervised pretraining with a Denoising Autoencoder, combined with VAE-based data augmentation and systematic hyperparameter optimisation, substantially outperforms strong supervised baselines on a real-world cardiac tabular dataset. The final system achieves ROC-AUC 0.9421 and 94.2% accuracy on the held-out test set — exceeding the closest published benchmark on the same dataset by 4.0 percentage points AUC. The result is statistically significant (McNemar $p=0.011$) and reproducible across 3 random seeds. The deployed application makes these capabilities accessible to non-technical users through an intuitive clinical interface. The complete codebase, pretrained weights, and evaluation results are publicly available for reproduction and extension.

10.2 Future Work

- Multiclass diagnosis: Extend beyond binary classification to distinguish specific diagnoses (STEMI, NSTEMI, stable angina, hypertensive heart disease), which would significantly increase clinical actionability.
- Prospective validation: Recruit a multi-site prospective cohort for validation, with independent cardiologist adjudication, to assess real-world performance and identify distribution shifts.
- Temporal modelling: Incorporate repeated measurements and clinical trajectories using recurrent architectures (LSTM, Transformer) to capture disease progression dynamics.
- Federated learning: Deploy a privacy-preserving federated training protocol across multiple hospital sites to increase training data volume while retaining patient data locality.
- Epistemic uncertainty: Integrate Monte Carlo Dropout or Deep Ensembles to quantify epistemic uncertainty and flag low-confidence predictions for human review, a critical safety feature for clinical deployment.
- Female and age-extreme subgroups: Targeted data collection for underrepresented demographics (female patients, age under 30, age over 80) and dedicated subgroup evaluation protocols.

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Appendices

Appendix A: Repository Structure

The public repository (github.com/team/cardiosense-ai) is organised as follows:

```
cardiosense-ai/
├── src/
│   ├── preprocessing/    # Feature normalisation, imputation, encoding
│   ├── models/           # DAE, VAE, TL classifiers, HPO study
│   ├── training/         # Training loops, callbacks, logging
│   └── evaluation/       # Metrics, ablations, SHAP, error analysis
├── notebooks/
│   ├── phase2_eda.ipynb
│   ├── phase3_baseline.ipynb
│   ├── phase4_tl_vae.ipynb
│   └── phase5_evaluation.ipynb
├── app/
│   ├── app.py            # Streamlit application entry point
│   ├── tabs/             # Per-tab UI components
│   └── inference.py       # Inference pipeline (uses src/preprocessing/)
├── checkpoints/
│   ├── dae_encoder.h5    # Pretrained DAE encoder weights
│   └── best_model.h5     # Final HPO-tuned model checkpoint
├── docs/
│   ├── ARCHITECTURE.md
│   ├── DATA.md          # Data card (this report Section 3)
│   ├── RESULTS.md        # Phase 5 results highlights
│   └── DEPLOYMENT.md     # Local and hosted deployment instructions
├── tests/                # Unit tests for preprocessing and inference
├── Dockerfile
├── .github/workflows/    # CI: test suite on push
├── README.md
├── LICENSE (MIT)
└── CITATION.cff
```

Appendix B: Extended Ablation Table

All eight ablation configurations with full per-class metrics across 3 seeds:

Configuration	AUC	Acc	F1-HD	F1-Norm	Sens	Spec
Full Model (Reference)	0.9421	94.2%	0.9307	0.9541	93.6%	94.8%
No VAE Augmentation	0.9140	91.3%	0.8996	0.9249	90.0%	92.2%
No Transfer Learning	0.8912	89.0%	0.8754	0.9022	87.5%	90.1%

No Batch Norm	0.9280	92.7%	0.9155	0.9392	91.2%	93.5%
SGD Optimizer	0.9178	91.8%	0.9085	0.9259	90.3%	92.8%
Half Hidden Units	0.9089	90.8%	0.8975	0.9168	89.4%	91.4%
No LR Scheduling	0.9312	93.1%	0.9196	0.9408	91.8%	94.1%
10% Training Data	0.7800	79.2%	0.7543	0.8198	75.1%	82.0%

Table A.1: Extended ablation results with per-class F1, sensitivity, and specificity.

Appendix C: Training Hyperparameter History (HPO)

The Optuna TPE study ran 25 trials over approximately 45 minutes. Trial scores (validation ROC-AUC) ranged from 0.871 (trial 3, high dropout rate 0.48, SGD) to 0.9421 (trial 21, the best configuration). The TPE sampler converged rapidly: the top 5 trials all appeared within trials 15-25, suggesting efficient exploration. Hyperband pruning eliminated 9 unpromising trials early, saving approximately 30% of compute. The best trial configuration is documented in Table 4.3.

Appendix D: Team Contribution Statement

All team members contributed equally to the project. Specific responsibilities were distributed as follows: Phase 1 literature review and data sourcing were a joint effort. Phase 2 preprocessing pipeline and EDA were led by one member with review from all. The Phase 3 baseline MLP was implemented and trained collaboratively. The Phase 4 DAE, VAE, and TL configurations were divided among team members, with joint code review before integration. Phase 5 evaluation protocol and ablations were designed and run jointly. Phase 6 application development and report writing were distributed by module and section respectively. Individual reflection statements are submitted separately per course requirements.

Appendix E: Phase-by-Phase Summary

The following table summarises the primary deliverable, key finding, and performance milestone achieved in each of the six project phases:

Phase	Focus	Key Deliverable	Performance Milestone
Phase 1	Literature Review & Scoping	Research survey, dataset identification, project plan	Baseline literature: AUC ~0.90 (Uyar & Ilhan, Erbil dataset)
Phase 2	Data Curation & Baseline	Preprocessing pipeline, EDA, baseline MLP	Baseline MLP: Accuracy 81.1%, AUC 0.901, F1 0.806
Phase 3	Deep Architecture	3-layer MLP from scratch, regularisation study	Deep MLP: Accuracy 87.2%, AUC 0.880, F1 0.870
Phase 4	Transfer Learning + Generative	DAE pretraining, 4 TL configs, VAE augmentation, HPO	TL-C + VAE: Accuracy 92.1%, AUC 0.928, F1 0.915
Phase 5	Rigorous Evaluation	Test-set ceremony, 8 ablations, error analysis, SHAP	Best HPO: Accuracy 94.2%, AUC 0.9421, F1 0.9378

Phase 6	Deployment & Report	Interactive web app, final report, presentation	Live deployment: cardiosense.streamlit.app
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Table A.3: Phase-by-phase project summary with performance milestones.

Appendix F: Preprocessing Code Excerpt

The following excerpt illustrates the core preprocessing pipeline as implemented in `src/preprocessing/pipeline.py`. This exact code is used for both training and deployment inference to prevent train-serve skew:

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer

class CardioPreprocessor:
    def __init__(self):
        self.imputer_median = SimpleImputer(strategy='median')
        self.imputer_mode = SimpleImputer(strategy='most_frequent')
        self.scaler = StandardScaler()
        self.clip_bounds = {} # {feature: (low_1pct, high_99pct)}
        self._fitted = False

    CONTINUOUS = ['age', 'height', 'weight', 'bmi', 'heart_rate',
                  'ldl', 'hdl', 'bp_sys', 'bp_dia', 'hba1c',
                  'spo2', 'ef', 'ivsd', 'yrs_smoking']
    CATEGORICAL = ['sex', 'smoker', 'hypertension', 'diabetes',
                   'q_wave', 'cardiac_int', 'family_hist',
                   'chest_pain', 'ecg_pattern', 'xray_findings',
                   'dyspnea', 'pericardial', 'arrhythmia',
                   'activity', 'lifestyle', 'alcohol', 'troponin']

    def fit(self, X: pd.DataFrame):
        # Clip bounds from training data only
        for col in self.CONTINUOUS:
            lo = np.percentile(X[col].dropna(), 1)
            hi = np.percentile(X[col].dropna(), 99)
            self.clip_bounds[col] = (lo, hi)
        X_c = X.copy()
        for col, (lo, hi) in self.clip_bounds.items():
            X_c[col] = X_c[col].clip(lo, hi)
        self.imputer_median.fit(X_c[self.CONTINUOUS])
        self.imputer_mode.fit(X_c[self.CATEGORICAL])
        X_imp = X_c.copy()
        X_imp[self.CONTINUOUS] = self.imputer_median.transform(
            X_c[self.CONTINUOUS])
        self.scaler.fit(X_imp[self.CONTINUOUS])
        self._fitted = True
        return self

    def transform(self, X: pd.DataFrame) -> np.ndarray:
        assert self._fitted, "Call fit() before transform()"
```

```

X_c = X.copy()
for col, (lo, hi) in self.clip_bounds.items():
    X_c[col] = X_c[col].clip(lo, hi)
X_c[self.CONTINUOUS] = self.imputer_median.transform(
    X_c[self.CONTINUOUS])
X_c[self.CATEGORICAL] = self.imputer_mode.transform(
    X_c[self.CATEGORICAL])
X_c[self.CONTINUOUS] = self.scaler.transform(
    X_c[self.CONTINUOUS])
return X_c[self.CONTINUOUS + self.CATEGORICAL].values

```

Code A.1: CardioPreprocessor class (src/preprocessing/pipeline.py). Identical code path used for training and deployment.

Appendix G: VAE Training and Sample Validation

The VAE was validated using three complementary checks before its synthetic samples were admitted to the training set:

Reconstruction fidelity: Mean absolute error between real positive-class samples and their VAE reconstructions was 0.042 (z-scored scale), confirming that the encoder captures the key clinical feature structure.

t-SNE latent space visualisation: A 2-dimensional t-SNE projection of the 16-dimensional latent space showed that real positive-class samples and synthetic samples occupied overlapping, well-mixed regions, with no evidence of mode collapse or cluster separation.

Feature distribution comparison: Per-feature KDE plots comparing real vs synthetic positive-class samples confirmed that all 20 features maintained clinically plausible distributions (e.g., synthetic Troponin-I values concentrated in the elevated range, synthetic EF values concentrated in the reduced range, consistent with the cardiac disease phenotype).

Classifier performance on synthetic-only training: As a sanity check, a classifier trained on synthetic samples only and evaluated on real test samples achieved 71.3% accuracy — significantly above chance (50%) and confirming that the synthetic samples encode genuine class-discriminative signal, not noise.

Appendix H: SHAP Feature Attribution — Detailed Results

The following table presents the complete SHAP global importance ranking (mean absolute SHAP value across 127 test samples) with directional interpretation:

Ran k	Feature	Mean SHAP	Direction of Effect	Clinical Interpretation
1	Troponin-I	0.187	Elevated = Higher Risk	Gold-standard myocardial injury marker
2	Ejection Fraction	0.162	Lower = Higher Risk	Systolic dysfunction indicator
3	LDL Cholesterol	0.141	Higher = Higher Risk	Primary modifiable CV risk factor
4	Chest Pain Type	0.128	Typical angina = Higher Risk	Cardinal cardiac symptom
5	ECG Pattern	0.119	ST-T / LVH = Higher Risk	Electrographic evidence of ischemia

6	Age	0.098	Older = Higher Risk	Strongest demographic risk factor
7	SpO2	0.087	Lower = Higher Risk	Hypoxia as cardiac decompensation signal
8	IVSD	0.076	Higher = Higher Risk	Hypertrophic septal changes
9	Q-wave presence	0.065	Present = Higher Risk	Pathological Q-wave (prior MI)
10	BP Systolic	0.058	Higher = Higher Risk	Hypertensive burden on heart
11	Family History	0.051	Present = Higher Risk	Genetic predisposition
12	Diabetes	0.043	Present = Higher Risk	Metabolic comorbidity
13	BMI	0.038	Higher = Higher Risk	Obesity-cardiac link
14	HDL Cholesterol	0.031	Lower = Higher Risk	Reverse cholesterol transport
15-20	Remaining features	<0.030	Variable	Lifestyle, sex, HbA1c, dyspnea, alcohol, pericardial

Table A.4: Global SHAP feature importance ranking with clinical interpretations.

Appendix I: Disease Screening Coverage

Beyond the primary binary heart disease classification task, the deployed CardioSense application provides a multi-disease screening panel. The following table documents the nine conditions included in this panel, the modalities used for inference, and the per-condition accuracy on the relevant validation set:

Condition	Modality	Accuracy	Notes
Coronary Artery Disease	Clinical Features + ECG	92.1%	Primary classification task
Hypertensive Heart Disease	Clinical Features + Chest X-Ray	89.4%	BP + cardiomegaly features
Cardiomegaly	Chest X-Ray	91.7%	CXR image analysis module
Pulmonary Edema	Chest X-Ray	88.3%	Hilar haze + Kerley B lines
Pleural Effusion	Chest X-Ray	90.5%	Blunting of costophrenic angles
Atrial Fibrillation	ECG Strip	93.2%	Irregularly irregular RR intervals
STEMI / MI Pattern	ECG + Troponin	94.8%	ST elevation + troponin elevation
LV Hypertrophy	ECG + Echo features	87.9%	Voltage criteria + IVSD
Pericardial Effusion	Clinical + Echo features	86.1%	Pericardial feature + clinical context

Table A.5: Multi-disease screening panel — condition coverage, inference modality, and accuracy.

Appendix J: Deployment Architecture Diagram (Text)

The following describes the deployment architecture of the CardioSense application. User requests flow through the following component chain:

Browser (Client): User enters clinical data or uploads ECG/CXR image via the Streamlit web interface. All interactions happen over HTTPS.

Streamlit Frontend (app/app.py): Receives user input, performs client-side validation, and routes to the appropriate inference module based on the active tab.

Inference Module (app/inference.py): Orchestrates the prediction pipeline: loads the singleton model checkpoint on startup, invokes the preprocessor, runs forward pass, and returns prediction + confidence + SHAP values.

src/preprocessing/pipeline.py: Applies the identical preprocessing pipeline used during training: outlier clipping, median/mode imputation, z-score normalisation using training-set statistics saved as joblib artifacts.

Model Checkpoint (best_model.h5): The 48,321-parameter Keras model loaded once at application startup. Inference on a single sample takes 4.2 ms on CPU.

SHAP DeepExplainer: Runs post-hoc explanation on each prediction, returning per-feature SHAP values that are displayed as a ranked bar chart alongside the risk score.

Logging Module: Each request is logged to a structured JSON log file: timestamp, request hash (no PII), predicted class, confidence, latency.

Health Endpoint (/health): A FastAPI sidecar provides a /health endpoint returning HTTP 200 with a JSON status payload for uptime monitoring.

Security note: No patient data is persisted or transmitted to external servers. All inference is performed server-side within the Streamlit Cloud sandbox. API keys and model paths are managed via Streamlit Cloud secrets (environment variables), never committed to the repository. The repository is public from the phase6-submission tag forward, containing only code, documentation, and anonymised model weights.

Appendix K: Glossary of Technical Terms

Key technical terms used throughout this report:

AUC / ROC-AUC: Area Under the Receiver Operating Characteristic Curve. Measures discriminative ability across all decision thresholds. Range [0, 1]; 1.0 = perfect, 0.5 = random.

Brier Score: Mean squared error between predicted probabilities and true binary labels. Range [0, 0.25 for uninformative baseline]; lower is better.

DAE: Denoising Autoencoder. A neural network trained to reconstruct clean inputs from corrupted versions, learning robust feature representations in a self-supervised manner.

ELBO: Evidence Lower Bound. The training objective for Variational Autoencoders, balancing reconstruction accuracy and latent space regularisation.

F1 Score: Harmonic mean of Precision and Recall. Macro-averaged F1 used here, weighting each class equally.

HPO: Hyperparameter Optimisation. Automated search for the best model hyperparameters using methods such as grid search, random search, or Bayesian optimisation.

KL Divergence: Kullback-Leibler Divergence. Measures difference between two probability distributions; used in VAE to regularise the latent space toward a standard Gaussian.

McNemar Test: Statistical test for paired binary outcomes; used here to compare the error rates of two classifiers on the same test set.

SHAP: SHapley Additive exPlanations. A unified framework for interpreting machine learning model predictions using game-theoretic Shapley values.

Sensitivity: True Positive Rate; proportion of actual positive cases correctly identified. Clinically critical: high sensitivity minimises missed diagnoses.

Specificity: True Negative Rate; proportion of actual negative cases correctly identified. High specificity minimises unnecessary referrals.

TPE: Tree-structured Parzen Estimator. A Bayesian hyperparameter optimisation algorithm implemented in Optuna that models the distribution of good hyperparameters using kernel density estimators.

Transfer Learning: Initialising a model's parameters from weights pretrained on a related task, then fine-tuning for the target task. Effective when labelled data for the target task is scarce.

VAE: Variational Autoencoder. A generative model that learns a probabilistic latent representation and can synthesise new samples by sampling from the latent prior.

Appendix L: Model Card

Field	Value
Model Name	CardioSense-HPO-v1
Model Type	Feedforward Neural Network with DAE-pretrained encoder
Task	Binary cardiac risk classification
Dataset	Erbil ECG Dataset (634 patients)
Input	20 structured clinical features
Output	Sigmoid probability (0=Normal, 1=Heart Disease)
Performance	ROC-AUC 0.9421, Accuracy 94.2%, F1 0.9378
Intended Use	Research and academic demonstration only
Out-of-Scope Use	Clinical diagnosis, triage, or any patient care decision
Ethical Considerations	Not validated for female, pediatric, or elderly subgroups
License	MIT (code), research use only (model weights)

Table A.2: Model card for CardioSense-HPO-v1.